2026 INTRODUCTION TO ML & DL

MIDTERM PROJECT COMPETITION

2026.10.21



DATASET AND TASK

How badly was this building damaged?

Dataset

Real-world post-earthquake building damage survey

Task: Multi-class Classification

根據建物震前的結構與使用特徵，預測它在地震中的損壞程度

3

damage levels to predict

15,000

buildings in train.csv

5,100

buildings in test.csv

LABEL

Three damage levels

LEVEL

1

輕微

Negligible to moderate damage, cracks in walls

LEVEL

2

中度到嚴重

Substantial to very heavy damage, structural damage

LEVEL

3

倒塌

Destruction, total or near-total collapse

label：損壞程度，整數 1–3

DATA ATTRIBUTE INFORMATION

5 named features, 233 anonymised ones

train.csv (with label) 、test.csv (no label) , both files have an id column.

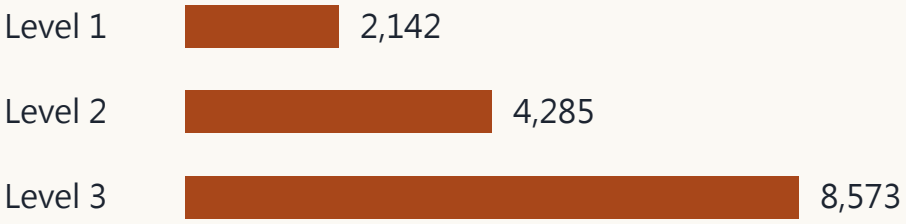
Named features | Anonymized features

structure _age	Storey _count	footprint _area	Roofing _kind	Terrain _slope	f_1	f_2	…	f_232	f_233	label
int	int	int	category	category	float or int	float or int	…	float or int	float or int	1 or 2 or 3

CLASS DISTRIBUTION

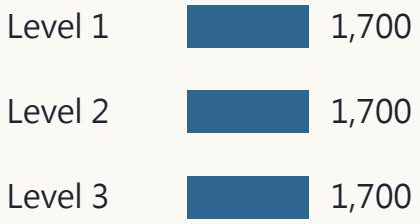
Imbalanced data

Train data · 15,000



→ 約 1 : 2 : 4

Test data · 5,100



→ 1 : 1 : 1

KAGGLE

Join the competition

01 Create a Kaggle account (register ONLY with your NTU COOL login email)

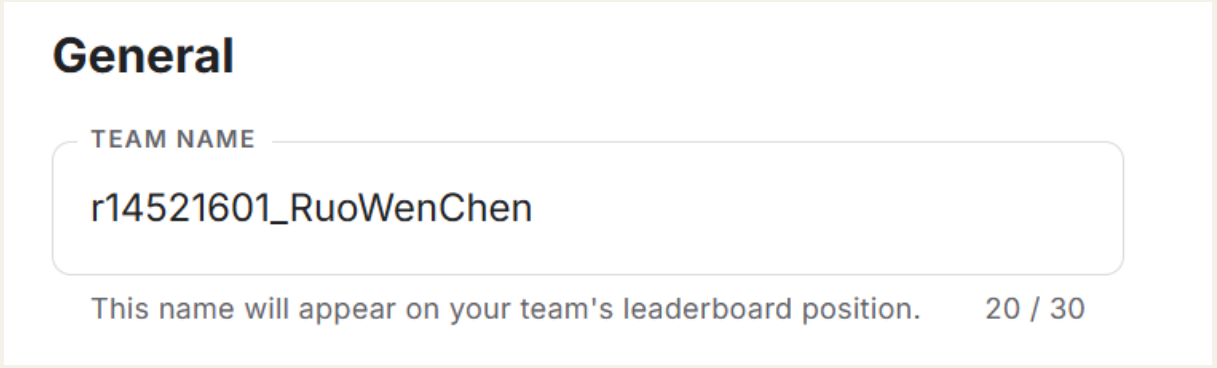
02 Join the competition ： 2026 ML&DL Midterm Project Competition

03 Team Name：

台大學生：學號_任意名稱（ ex: r14521000_XX ）

台科大學生：ntust_學號_任意名稱

台師大學生：ntnu_學號_任意名稱

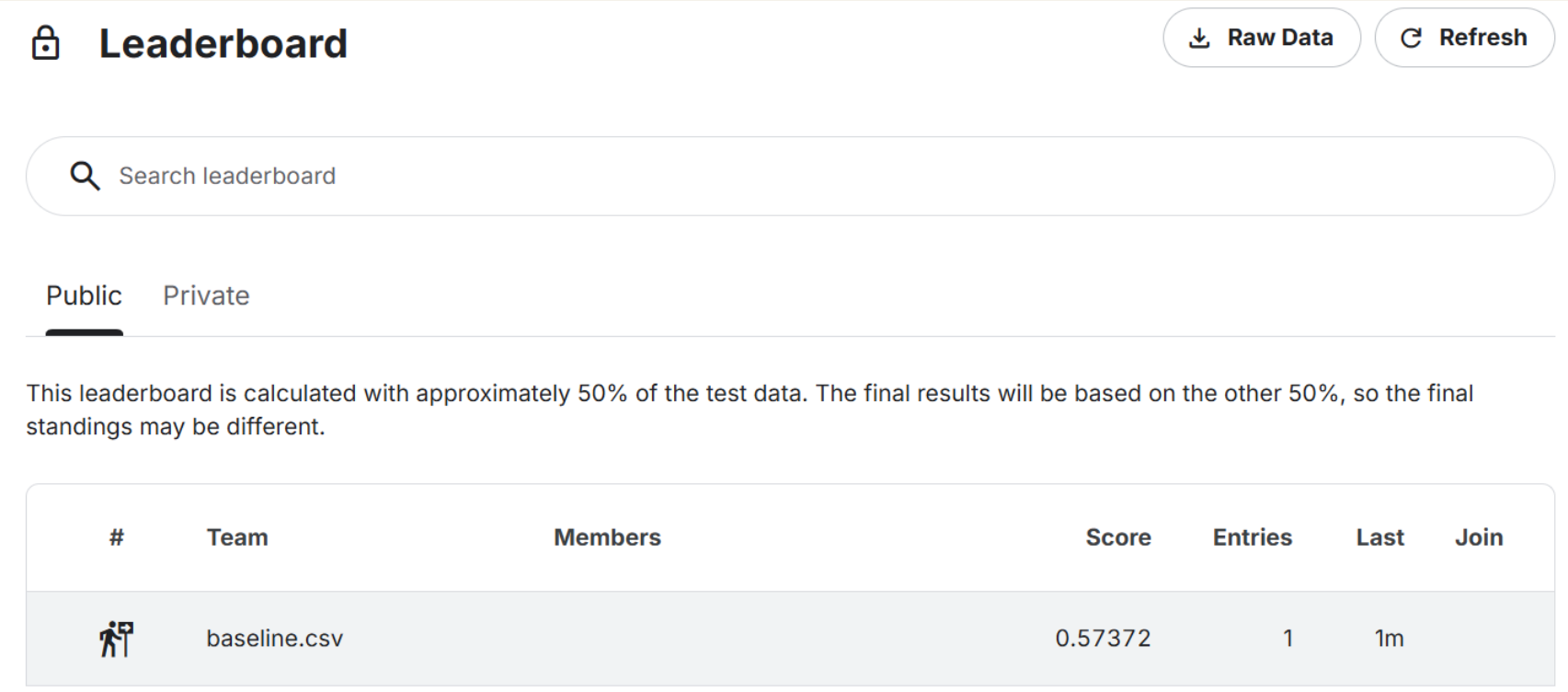


LEADERBOARD

Leaderboard

Simple baseline

- Public: 0.57372
- Private: published after deadline



SUBMISSION

Submission

- Predict all 5100 samples in test.csv
- You can submit **5 times every day**
- The daily submission limit resets every day at UTC 00:00 **(08:00 Taiwan time)**
- Upload format: .csv
- The first row must be id, label, and the predictions start from the second row.
- Please use period to seperate id and label in each row.
- Evaluation: accuracy

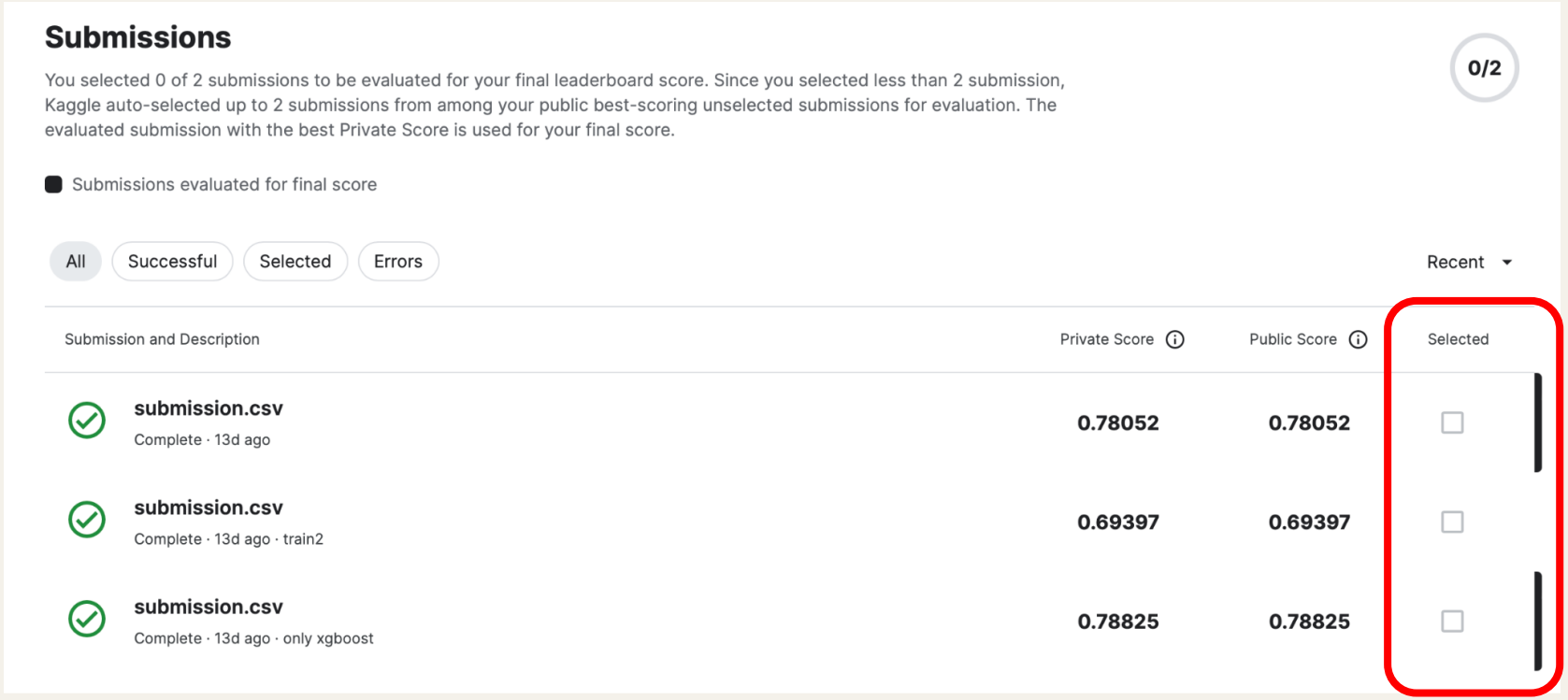
sample_submission.csv

id	label
1	3
2	3
3	3
4	3
5	3
6	3
7	3
8	3
9	3
10	3
11	3
12	3

SUBMISSION

Selection

- Select 2 submissions for final ranking.
- The evaluated submission with the best Private Score is used for your final score.



GRADING · 30% OF THE COURSE

How the midterm competition is scored

Item	Weight	Detail
Milestone	3%	Submit one output on Kaggle before leaving class
Baseline	55%	public 35% · private 20%
Rank	30%	public 15% · private 15%；過 baseline 者依名次給分，第 1 名滿分
Top 5	10%	public 5% · private 5%；第 1 名 5%、第 2 名 4% …
Report + Code	2%	Report 1% · Code 1%

Ex：101 人過 baseline，你是 leaderboard 上第 25 名，Rank 得 ((100 – 24) / 100) × 15 = 11.4 分

11/03 22:00 公布受邀分享名單，被邀請沒出現者，斟酌扣分

Deadline

- Kaggle Deadline: **11/03 (Tuesday) 21:00**
- Report &Code Deadline: **11/03 (Tuesday) 23:59**
  - Report: template (in pdf)
  - Code: .ipynb, .py
  - Submit: mid_r14521601.zip (report + code)

  - The list of students invited to present will be announced on 11/03 at 22:00

TODAY

Milestone

- Submit one output on Kaggle (other than the sample submission) before leaving class.
- Try to train an ML model on the training data, make a prediction, generate a submission file, and submit it on Kaggle.
- Show your submission to the TA before leaving.
- You'll get the 3% milestone score.